Akhand Raj Singh

Full Stack Developer (React.js, Next.js, Node.js, MongoDB) | Software Developer
Noida, Uttar Pradesh | 7379660997 | akhandrajsingh30@gmail.com
linkedin.com/in/akhand-raj-singh | github.com/Fit-akhand

Profile Summary

Final-year B.Tech Computer Science student (2027 batch) with strong fundamentals in C++, JavaScript, Python, OOP and DSA (260+ LeetCode problems). Hands-on experience building, debugging and deploying full-stack web apps with databases and REST APIs. Built two live multi-role products, took part in two hackathons, and am keen to learn and contribute in a fast-paced engineering team.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C++
Web: HTML5, CSS3, React.js, Next.js (App Router), Tailwind CSS, Node.js, Express.js, Flask
Databases & APIs: MongoDB, Mongoose, Atlas, REST API development, Postman
Fundamentals: Data Structures & Algorithms, OOP, MVC & Layered Architecture, Debugging
Auth & Security: JWT, HTTP-only cookies, Bcrypt, OTP verification, RBAC, Zod validation
Tools & Deployment: Git, GitHub, Postman, Cloudinary, Vercel
AI (exposure): LLM integration, RAG, NLP, vector databases

Projects

NovaCart – Multi-Vendor E-Commerce Marketplace Live Demo
Next.js, React.js, Node.js, MongoDB, Mongoose, JWT, Cloudinary, REST APIs

- Built and deployed a multi-vendor marketplace with separate **Customer, Seller and Admin dashboards**, using JWT authentication and role-based authorization.
- Developed product management (**inventory, pricing, discounts, images**) with seller-owned data isolation, plus a cart and order pipeline tracking payment, delivery status and seller-scoped revenue.
- Structured the app as **UI Components → Global State → API Routes → Auth Middleware → Mongoose → MongoDB**; integrated Cloudinary for image uploads.

The Hostel Yard – Hostel Management SaaS Live Demo
React.js, Node.js, Express.js, MongoDB, Mongoose, JWT, Bcrypt, Zod

- Built and deployed a production-oriented, multi-role platform for **Business Owners, Landlords, Property Managers, Staff and Tenants** across organizations and properties.
- Designed a scalable **Route → Controller → Service → Model** architecture with organization-level data isolation and permission-based RBAC across **5 roles**.
- Implemented secure authentication with **JWT, HTTP-only cookies, Bcrypt, OTP verification**, token versioning, and expiring hashed password-reset and invitation tokens.
- Modeled operations end to end (**Properties → Rooms → Beds → Tenants → Bookings → Rent → Payments**) with allocation-conflict checks, **Zod validation** and a reporting dashboard.

WellMind AI – AI-Powered Mental Wellness Support System Ongoing
React.js, TypeScript, Node.js, Express.js, MongoDB, Python, Flask, LLMs, RAG, NLP, Vector Database

- Developing a full-stack wellness platform with an **LLM-powered chatbot** that gives context-aware support and suggests self-care activities from a curated knowledge base.
- Implementing a **RAG pipeline**: document preprocessing, chunking, embedding generation, vector similarity search and context-grounded response generation.
- Building **REST APIs** for authentication, chat, mood and survey data, and integrating Python/Flask AI services with the Node.js backend, with emphasis on privacy and responsible AI safeguards.

Experience & Hackathons

Backend Developer – Smart India Hackathon (SIH) Sept – Oct 2025
Problem Statement 25010: Smart Crop Advisory System for Small and Marginal Farmers

- Contributed to **backend architecture**, designing the server-side structure and core application flow in a team.
- Implemented secure **authentication and authorization**, and designed **database connections and data models** for reliable storage and retrieval.

Frontend Developer – Tech Hackathon Nov – Dec 2024
JSS Academy of Technical Education, Noida, Uttar Pradesh

- Developed responsive UI components with **React.js, Tailwind CSS and React Router** for a **service-booking platform**, collaborating with teammates under hackathon deadlines.

Education

JSS Academy of Technical Education, Noida, Uttar Pradesh Expected Graduation: 2027
Bachelor of Technology in Computer Science

ND DAV Public School, Ayodhya, Uttar Pradesh
Class XII (CBSE), 2022 | Class X (CBSE), 2020